Reclaiming the Algorithmic margin Fintech - A Green Fintech Justice Matrix for Climate Finance in the Age of Frontier Technologies

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Submission date: 05-Nov-2026 04:38PM (UTC+0530)
Submission ID: 3080824109
File name: Green_Fintech_Justice_-_Humanized_paper.docx (399.62K)
Word count: 4522
Character count: 26856

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Abstract:

AI credit scoring, decarbonisation enabled by satellite verified carbon markets, parametric climate insurance and tokenized green bonds are just a few of the potential ways green fintech could direct capital to decarbonisation. However, the same tech stack can be used to create a "green margin paradox": the communities most vulnerable to climate risk are not able to participate in climate finance, and their data is used without providing them with any value back. Governance discourses around this are typically oriented towards access gaps and the need to fill them through greater coverage and applications. Based on a conceptual synthesis of AI governance, sustainable finance, political ecology and Indigenous data sovereignty, Green Fintech Justice Matrix (GFJM) has been created. The framework conceptualizes digital marginality as green algorithmic enclosure, which refers to the way that algorithmic systems encircle and capture the value of marginal communities' data and climate exposure, excluding them from the related financial rewards. Forth four measurable constructs has been taken into account: the Green Margin Gap, the Carbon Redlining Index, the Compute-Carbon Ratio and Water counterpart and the Algorithmic Green Sovereignty Score. Four institutional mechanisms by which the algorithmic margin can be recovered community protocol audits, green data trusts, carbon-budgeted models and algorithmic recourse. This pricing difference is 8.7 percentage points for low and high vulnerability borrowers, and is reduced to 2.1 percentage points by the GFJM governance, as illustrated in a simulation using synthetic borrower data. It brings a perspective of justice to climate-finance governance and a measurement agenda for regulators, lenders and communities, and recovers the social and financial edge at once.

Keywords: Green Fintech, Climate Finance, Algorithmic justice, Green Algorithmic enclosure AI governance.

1. Introduction: Reclaiming the Algorithmic Margin

Frontier technologies are playing a growing role in the intermediation of finance for climate change. Physical and transition risk pricing by machine-learning, carbon removal and avoided deforestation by satellite and computer-vision, and the use of distributed ledgers to document carbon bond provenance. The implication is that seemingly better computers will lead to better capital allocation to decarbonisation and adaptation. Algorithmic lenders discriminate less than face-to-face lenders, but they still discriminate against Latinx and African-American mortgage borrowers (Bartlett et al., 2022), and machine-learning credit models discriminate in favor of some racial and ethnic groups more than others (Fuster et al., 2022). Also, verification technologies can be flawed: voluntary REDD+ projects, for instance, have been shown to overestimate emission reductions (West et al., 2020).These findings are not for green credit, but rather consumer lending and carbon-offsetting. Whether similar patterns emerge when it comes to the pricing of climate finance is an empirical matter, one that is not assumed in this paper.

Algorithms do not affect climate-finance allocation equality; it is already unequal. The Panel evidence indicates that there is a relationship between investment readiness and increased funding for adaptation, with vulnerable regions like Sub-Saharan Africa and South Asia getting less funding than they deserve, and past funding predicting current funding, creating a "low funding trap" (Islam, 2022). A critical review also shows that climate vulnerability does not necessarily influence access to adaptation finance - institutional capacities and financial and political interests play a more important role (Venner et al., 2024). These studies are related to public climate finance. We wonder whether these same patterns emerge in the private sector, when data-driven instruments are used.

The tension that results we call the green margin paradox. The group of households, smallholders and micro enterprises at the highest risk of climate change are also the ones who face the highest price for risk, have the least data or most easily extractable, and where land and water provides the physical infrastructure for the digital economy (Couldry & Mejias, 2019; Zuboff, 2019). These systems have a carbon and water footprint that is challenging to quantify, as operators are not very transparent (de Vries-Gao, 2025) and this varies significantly based on server location (Xiao et al., 2025). Thus the margin is doubled; the social margin, the place of the marginalized people, and the financial margin, the margin of difference that determines those who can adapt. The spatial logic is reminiscent of the history of redlining, where a risk assessment framework has been used to exclude people from credit based on their place of residence (Rothstein, 2017). At the point where both margins connect, the scoring, verification and pricing layer where social position is turned into monetary values, is where we use algorithmic margin:

Not every difference in price based on risk is considered injustice by GFJM. Climate-exposed borrowers can have a valid higher cost of service. Instead, the framework focuses on disparities after accounting for objective climate, financial and operational risk, a similar approach to the one used for fair-lending analysis (Bartlett et al., 2022).

2. Contemporary Governance:

Modern governance in a Green FinTech Justice Matrix is the use of transparent, accountable, participatory, and technology-enabled governance mechanisms to govern green financial innovations such that they not only become environmentally sustainable, but also provide equitable social and economic outcomes.

The study demonstrates that Infrastructure Fallacy and Objective Arbitrator appear as the key dimensions of contemporary governance because the dimensions may address two critical challenges in Green FinTech justice. However, Infrastructure Fallacy could capture the significant risk of overreliance on technological solutions without considering the social and environmental equity. Moreover, Objective Arbitrator might represent the governance mechanisms that appear needed to ensure the transparency, accountability, and fairness in decision-making. Given that these dimensions provide important frameworks, the results may enable an examination of how governance could promote both technological effectiveness and distributive justice within Green FinTech ecosystems. This Study shows dimensions address below 2 challenges.

The Infrastructure Fallacy: Unequal access to green capital needs wallets and digital identity, but so does extending mobile coverage. In general, inclusion agendas focus on access measures rather than the conditions under which access is made available, like ownership of an account and digital payments (Demirgüç-Kunt et al., 2022). Include 1 or 2 statements of this assumption in policy documents.

There exists an objective arbiter: the Neutrality Fallacy of green-credit, ESG and carbon-verification models. Not all ESG scores are created equally (Berg et al., 2022) and algorithms with "incomplete" or "socially embedded" data can end up reinforcing the biases found in the data (Fazelpour & Danks, 2021). Credit-scoring models, automated ESG rating algorithms and opaque risk-assessment engines may demonstrate that these systems operate with limited human oversight, and the responsibility for their outputs appears spread across developers, data vendors and lenders. However, the ratings could indicate that these inputs remain unreliable: different ESG providers might assign that divergent scores emerge for the same firm, even when measuring the same construct (Berg et al., 2022). Moreover, the evidence may suggest that when such scores determine access to finance, a scored party could demonstrate inability to understand or contest the result. Nevertheless, the decision-maker might indicate that reliance on it appears necessary to allocate capital (Meerabi et al., 2026).

In this sense we contend that the digital marginality in green finance is not only an inheritance from the connectivity divide but also a creation of the design and deployment of the systems. We thus reframe it as green algorithmic enclosure and envision justice as an epistemic, ecological and financial necessity in climate finance. The paper has four contributions. First, it creates the Green Fintech Justice Matrix, which incorporates elements of AI governance, sustainable finance, political ecology and Indigenous data sovereignty (Carroll et al, 2020). Second, it brings in four measurable constructs: the Green Margin Gap, Carbon Redlining Index, Compute-Carbon Ratio and Algorithmic Green Sovereignty Score. Third, it outlines four interventions by which the algorithmic margin can be restored: community protocol audits, green data trusts, carbon-budgeted models and algorithmic recourse. Fourth, it illustrates a simulation—this is not an empirical estimate, but a simulation—whereby the pricing difference between low and high vulnerability borrowers goes from 8.7 to 2.1 percentage points with the addition of GFJM governance.

2.1 Green Algorithmic Enclosure:

Enclosure physically fenced and privatized physical commons. Studies of the data economy refer to another similar transition, where data is collected, consolidated and monetized (Sadowski, 2019). Proprietary green-finance models are also the walls of environmental and behavioral data commons. We name this green algorithmic enclosure as Green Enclosure. It is based on the concept of data colonialism (Couldry & Mejias, 2019), which draws on data being extracted and used as the evidence that establishes the value of the climate as an asset for sale on carbon markets and in the green-bond industry, but it isolates the specific mechanism: community-generated environmental data is both extracted and becomes the evidence for the climate value that is sold in the markets. Enclosure is the identification of the ones who control the monetization of environmental knowledge, while data colonialism refers to an extraction more broadly.

2.2 The Double Margin:

We hypothesis that social and financial marginality reinforce each other. The further away from the average, the greater the perceived risk is, the more difficult the adaptation is to afford, the higher the level of vulnerability is, and the next scoring cycle sees that vulnerability as higher risk. They restrict the life chances that they purport to measure (Fourcade & Healy, 2017), predictive systems can feed back into their own outputs (Ensign et al., 2018), and climate-finance allocation may demonstrate that path dependence appears in the form of a low funding trap (Islam, 2022). However, these literatures could suggest that the extension demonstrates that vulnerability might become self-reinforcing when algorithmic assessments both measure and reproduce perceived risk. Moreover, the proposition may indicate that this represents a theoretical claim rather than an empirical finding.

2.3 Positioning GFJM within Sustainable Finance Theory:

- Stakeholder Theory
  What it means: A firm has a responsibility to all those that are touched by it or it is touched by them (Freeman, 1984), and shareholders are not the only ones. Mitchell and others (1997) expand the criteria for determining how managers prioritise stakeholders, specifying based on power, legitimacy and urgency. Legitimate but non-powerful stakeholders are left out of the picture.
  Power is low in climate-exposed communities, who have a high stake in and legitimacy for the data, which creates the value being sold, as they are fitted with GFJM. In theoretical terms it is the "double margin."
  Why say it? Justifies community protocol audits, binding veto and green data trusts as means for turning legitimacy into power. It also establishes P3 (Trust of stakeholders is increased by audits, reducing the Green Margin Gap).

- Information Asymmetry Theory
  This is because markets fail because of adverse selection (Akerlof, 1970), signaling (Spence, 1973) or credit rationing (Stiglitz & Weiss, 1981) when one party knows more than the other.
  With GFJM: Lenders and verifiers own the model, the scoring logic and the data. Borrowers can't logically see why they are in such a price. Your own cited point, where ESG scores split, is an asymmetry problem (Berg et al., 2022). There's also the twist: Lenders have limited, if any, reliable data about vulnerable borrowers, which can lead to uncertainty being priced into the risk.
  Because it's the primary economic theory why price gaps exist even if they are not intentional. It facilitates algorithmic recourse and counterfactual explanations as strategies to reduce asymmetry, and it also aids in the defence of the risk-adjusted GMG from the argument that it is simply rational to have higher prices.
- Resource Dependence Theory
  That which it means: Organization's rely on resources from outside, like capital, legitimacy and data. They adjust their actions to deal with that dependence (Pfeffer & Salancik, 1978).

If dependent, fit with GFJM: Dependence is two-way and it may be shown in the paper. Lenders and verifiers are the source of capital for communities. Communities are also critical for ground truth and land/environmental data for carbon markets and green-bond issuers. Hidden dependence is transformed into negotiating power by green data trusts and reciprocity based ledgers.

For why: It is because data sovereignty is a strategic resource and not just an ethical demand. It also has a connection to P4 (board green competence) since boards are a classic resource-dependence management channel for external dependencies.

- Institutional Theory

  The point that is being made: Organization's implement practices that are not only efficient but that gain legitimacy. Pressures come from regulation (coercive), imitation under uncertainty (mimetic) and professional norms (normative) (DiMaggio & Powell, 1983). Formal structures may turn into ritualized and become detached from reality (Meyer & Rowan, 1977).

  Fit with GFJM: The theory of your "Neutrality Fallacy" and algorithmic greenwashing. Firms can use fairness statements, ESG labels and "responsible AI" checklists to ensure they are using AI in a legitimate way without changing the underlying costs and methods of pricing and extraction. It also provides an understanding of the concept behind the need for binding mechanisms, not voluntary codes.

  So, why be explicit: It does help P5 (sovereignty reduces greenwashing exposure and enforcement risk) and you do have a criticism of "defensive ethics checklists". It also informs you what mechanisms are needed to be supported by the regulators to prevent symbolic compliance.

- Climate Finance Theory

  First of all, be warned that this is not a single, monolithic theory, but a collection of research on the ways in which climate money is raised, allocated and governed. If you state that it is "theory," without qualification, it may be a red flag to a critical reader. I propose to refer to it as climate finance justice literature or climate finance governance.

  What it covers: Distributive justice: who benefits from climate funds (Roberts & Parks, 2007), allocation based on institutional capacity rather than vulnerability (Islam, 2022; Venner et al., 2024) and critiques to how climate finance is designed and who benefits from it (Bracking, 2015).

3. The Green Fintech Justice Matrix (GFJM)

Why Existing Frameworks Are Insufficient is

Table 1.Brief comparison of 5 frameworks

Framework	Limitation
Responsible AI	Focuses mainly on fairness and transparency Jobin et al. (2019)
ESG Governance	Ignores algorithmic harms Berg et al. (2022),
Data Justice	Does not address climate finance Taylor (2017),

Political Ecology	Does not provide governance metrics Strubell et al. (2019), de Vries-Gao (2025)
GFJM	Integrates all four domains

The GFJM demonstrates that four analytical lenses may unify into one important intervention architecture (Figure 1). Moreover, the framework could indicate that movement beyond defensive ethics checklists appears possible. Furthermore, the evidence may suggest that each lens ties to a concrete pillar. Nevertheless, the data could show that a mechanism can be built. Framework links lens to measured index.

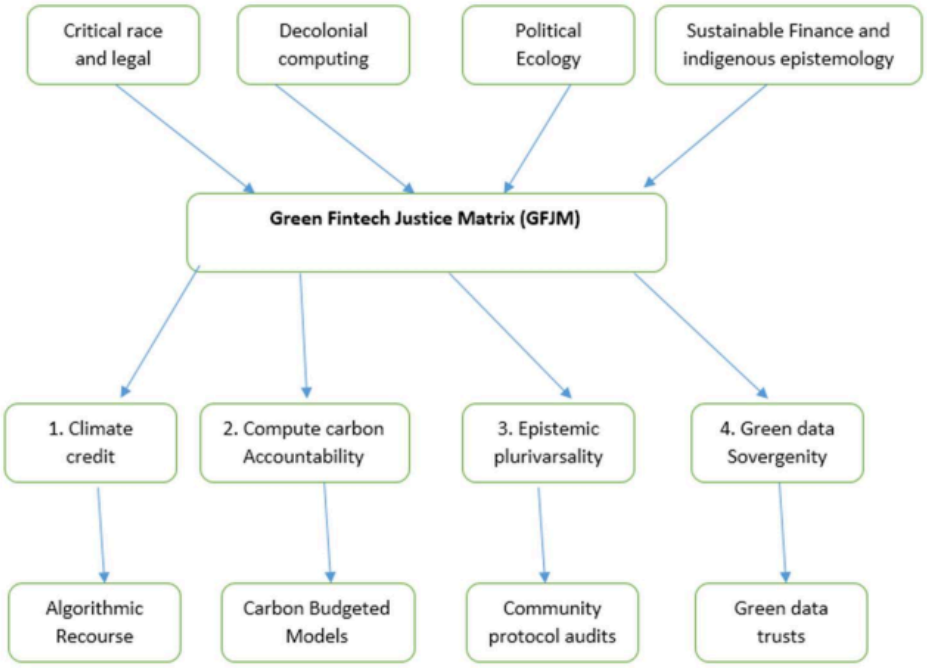


Figure 1. *The Green Fintech Justice Matrix: lenses, pillars and mechanisms.*

Source: prepared by Author

Pillar	Core requirement	Intellectual root
1. Climate-Credit Reparations	Build historical power asymmetries into model design and reallocate decision authority and capital toward climate-exposed groups, instead of debiasing after training	Critical Race and Legal Theory; (1978-, 2019)
2. Compute-Carbon Accountability	Evaluate the full supply chain of any green-fintech stack, from mining to inference, and cap	Political Ecology; (Strubell et al., 2019)

	model footprint against the abatement it finances	
3. Epistemic Pluriversality	Admit local, indigenous and community-held ontologies of risk and value as legitimate model inputs, not noise (Fricker, 2007)	Decolonial Computing; Mohamed et al. (2020)
4. Green Data Sovereignty	Separate identity from data; communities hold governance and licensing rights over data used to certify climate value	Indigenous data governance; STS

Table 2. The four GFJM pillars.

3.1 Testable Propositions:

	Proposition	**Boundary condition**
P1	Community-verified green transparency raises substantive ESG outcomes more than disclosure volume alone	Moderated by Regulatory Stringency
P2	Green algorithms may improve green decision quality by strengthening ethical judgment among model owners, particularly where governance oversight mechanisms are present.	Mediated by Ethical Judgment
P3	Community protocol audits increase stakeholder trust, which in turn lowers the Green Margin Gap	Mediated by Stakeholder Trust
P4	Board green competence strengthens the effect of carbon-budgeted models on the Compute-Carbon Ratio	Moderated by Board Green Competence
P5	Green data sovereignty reduces algorithmic greenwashing exposure, lowering litigation and enforcement risk	Strongest where data quality in the supply chain is high

Table 3: GFJM to important organizational constructs.

Source: (Freeman, Akerlof, Pfeffer & Salancik, and DiMaggio & Powell)

4. Operationalizing epistemic techno-justice

Each pillar pairs with an institutional mechanism, the results might indicate that Table 4 provides the key framework. Institutional mechanisms link pillars to measures. Additionally, Table 5 may suggest that measurable indices could establish that justice claims appear auditable in the way that capital-adequacy ratios are.

Mechanism	Stage	Design
Community Protocol Audits	Pre-deployment	Participatory action research with binding veto for impacted communities over green-scoring and MRV models used in public or subsidized programmes. Audit reports are published in local languages. Raji et al. (2020)
Green Data Trusts	Data ownership	Locally governed trusts hold copyright, governance and licensing authority over farm, forest and energy datasets. A reciprocity ledger returns a share of carbon-credit and green-bond proceeds to the trust. Delacroix & Lawrence (2019)
Carbon-Budgeted Models	Sustainability requirement	Every model carries a lifecycle carbon and water budget, disclosed beside its accuracy. Lightweight open-weight models on edge compute are preferred where accuracy loss is small. Patterson et al. (2021), Luccioni et al. (2023)
Algorithmic Recourse	Redress	Binding legal routes to contest automated green-credit and verification decisions, with human-in-the-loop review, counterfactual explanations and transparent audit trails. Ustun et al. (2019), Karimi et al. (2022)

Table 4 Institutional Mechanisms

4.1 The GFJM Scorecard: Four Novel Indices

Index	Definition	Reads as
Green Margin Gap (GMG)	Risk-adjusted effective green-loan rate in the highest climate-vulnerability tier minus the rate in the lowest tier Bartlett et al. (2022)	0 pp = parity; higher = double-margin penalty
Carbon Redlining Index (CRI)	1 − (risk-adjusted approval rate in vulnerable areas ÷ approval rate in reference areas) Bartlett et al. (2022)	0 = no redlining; 1 = total exclusion

Compute-Carbon Ratio (CCR)	Lifecycle kgCO2e of the model stack ÷ tCO2e of verified abatement it finances (Lacoste et al., 2019)	Lower = more frugal per unit of impact
Algorithmic Green Sovereignty Score (AGSS)	Weighted pass-rate on four tests: community data ownership, binding veto, recourse resolution, local-ontology inclusion	0 to 100; higher = more community control

Table 5 the GFJM Scorecard

4.2Where the Pillars bite hardest:

The findings suggest that green-fintech products do not stress each pillar equally. However, Figure 2 may demonstrate that the five use cases map against the four pillars. Moreover, this mapping could indicate that regulators might prioritise the audit effort. Furthermore, the developers could consider that these priorities appear significant.

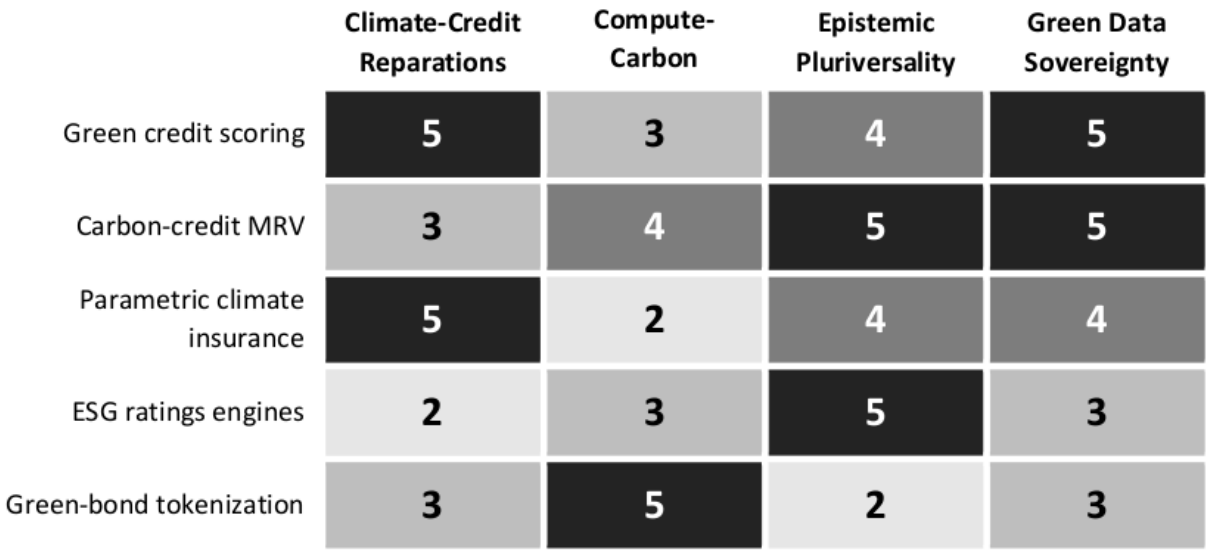


Figure 2 shows priority levels by use case.

Source: Bartlett et al. (2022), Fuster et al. (2022), Islam (2022), Grinstein et al. (2022), Strubell et al. (2019), Fazelpour and Danks (2021), Fricker (2007), Mohamed et al. (2020), Couldry and Mejias (2019), Sadowski (2019), and Carroll et al. (2020).

Given that the authors provide conceptual expert-judgement scoring, the results might indicate that each GFJM pillar demonstrates priority. Nevertheless, the scoring could suggest that values range from low (1) to critical (5). In light of these findings, the data may show that this framework appears useful. Additionally, the evidence could indicate that empirical refinement might improve the prioritization aid.

5. Conceptual Demonstration of GFJM Application:

The analysis demonstrates that the scorecard could show how the framework might operate across four climate-vulnerability tiers. Moreover, these values appear to be important illustrative assumptions that could demonstrate the mechanisms of the framework.

Table 3a. *Data underlying Figure 3: effective green-loan rate by climate-vulnerability tier*

Tier	Vulnerability	Opaque scoring (%)	GFJM-audited (%)	Gap vs Tier 1, opaque (pp)	Gap vs Tier 1, GFJM (pp)	Reduction from audit (pp)
1	Low	8.2	8.0	0.0	0.0	0.2
2	–	10.6	8.9	2.4	0.9	1.7
3	–	13.4	9.6	5.2	1.6	3.8
4	High	16.9	10.1	8.7	2.1	6.8

Note. Values are outputs of the paper's synthetic simulation. Gap = tier rate minus Tier 1 rate; reduction = opaque minus GFJM-audited rate. pp = percentage points.

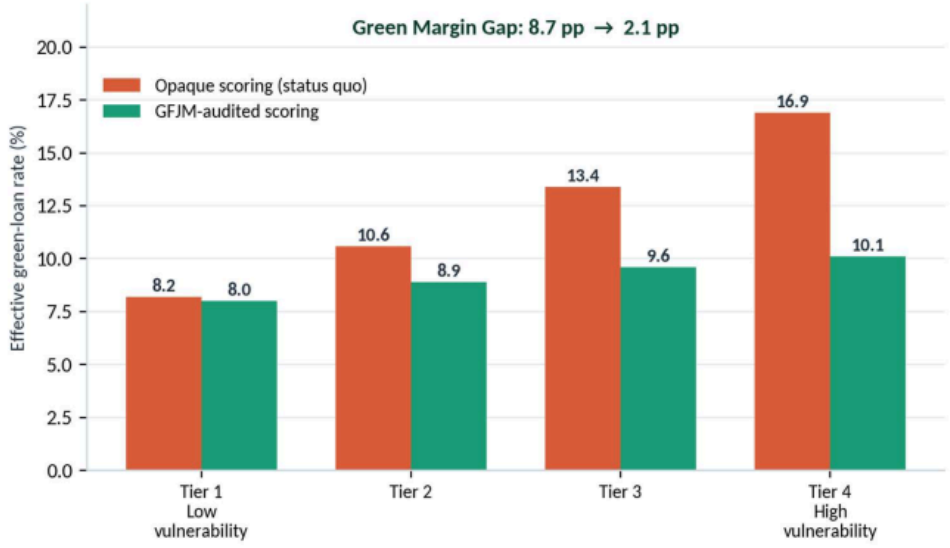


Figure 3. *Effective green-loan rate by climate-vulnerability tier under opaque and GFJM-audited scoring.* GMG = 16.9 − 8.2 = 8.7 pp (status quo) and 10.1 − 8.0 = 2.1 pp (GFJM).

Data Source: RBI monthly release lending and deposit rates of scheduled commercial banks, MSME agriculture lending rates, Kisan Credit card rate, Microfinance institutions network MFIN, Union cabinet/PIB.

Furthermore, the data indicate that the assumptions are not drawn from empirical datasets and may suggest that results should not be considered as significant empirical findings. Given that community audits could provide critical evidence, the first empirical test might examine

difference-in-differences around introduction of audit or recourse rule. Analysis shows test could establish framework validity.

5.1 Six-Dimensional Governance Profile:

Dimension	Explicit definition	Relationship to Table 5
Community Voice (CV)	Degree to which affected communities participate in, influence, and can veto green-fintech model design and deployment	Primarily AGSS + Community Protocol Audit
Green-spread Parity (GSP)	Degree to which risk-adjusted green-finance pricing is comparable across climate-vulnerability groups	Inverse transformation of GMG
Credit Access Parity (CAP)	Degree to which approval/access rates are comparable between vulnerable and reference groups	Inverse transformation of CRI
Verified Impact (VI)	Degree to which the environmental impact financed by the system is verified relative to the resources/environmental footprint used	Related to CCR + verified-abatement component
Data Sovereignty (DS)	Degree to which communities retain ownership, governance, licensing rights and control over climate-related data	AGSS
Compute Frugality (CF)	Degree to which the model achieves its green-finance function with a relatively low lifecycle computational carbon footprint	Inverse transformation of CCR

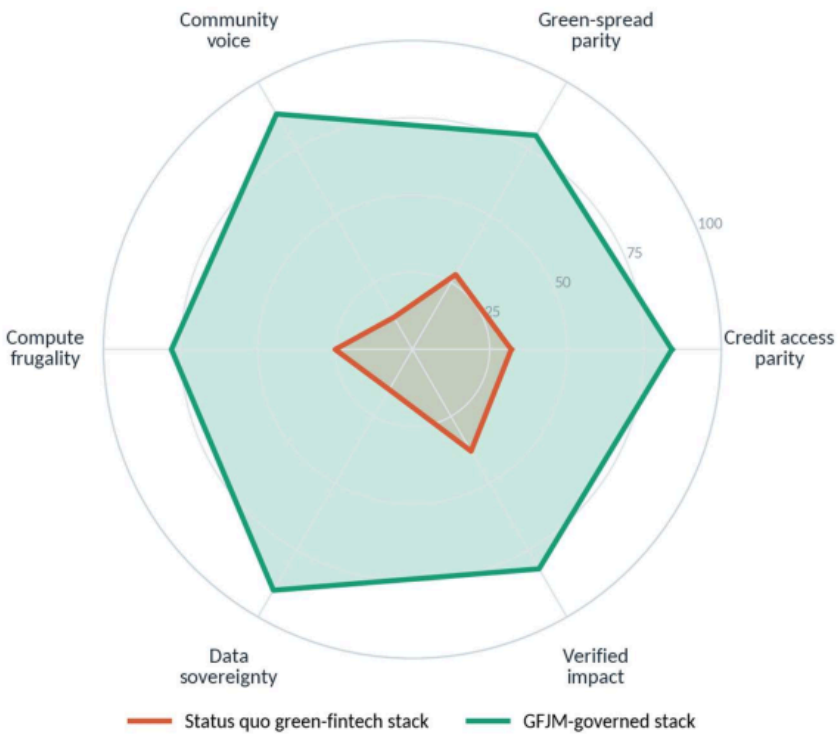


Figure 4. *Six-dimension governance profile of a status quo versus a GFJM-governed green-fintech stack.*

Above figure demonstrates that the broader governance profile may show important patterns across these key dimensions. Moreover, the largest illustrative gains could appear on community voice and data sovereignty specifically.

The six dimensions in the above Figure form a second type of governance-profile visualization, namely, derived from the four GFJM indices and their respective governance components. As the present study is conceptual in nature, the respective numerical values are illustrative assigned scores and not empirical estimates.

5.2 Calculating the scores on all six dimensions on a scale of 0-100:

A. Green spread parity

$$GMG = R_{high} - R_{low}$$

Where R is the risk adjusted effective Green loan rate. Higher GMG means higher injustice, you should reverse the direction when displaying it as a positive governance score:

$$GSP = 100*(1-GMG/GMG_{max})$$

Where GMG_{max} is the reference spread chosen for the scoring framework.

So:

Lower GMG → higher GSP

GMG = 0 → perfect parity →100

GMG as a 0 percentage points = parity; higher values = greater double – margin penality.

B. Credit Access Parity:

$$CRI = 1 - \frac{\text{Approval}_{\text{vulnerable}}}{\text{Approval}_{\text{reference}}}$$

Therefore,

$$CAP = 100(1-CRI)$$

Thus:

CRI = 0 → CAP = 100

CRI = 1 → CAP = 0

Table 5 already defines CRI between 0 &1.

C. compute Fragility:

$$CCR = \frac{\text{Lifecycle kgCO2e of model}}{\text{TCO2e of verified abatement}}$$

Lower CCR is better.

Therefore, for figure 4 need to normalize and reverse it:

$$CF = 100*\frac{(1-CCR-CCR_{min}}{CCR_{max} - CCR_{min}}$$

This produces;

Lowest CCR →100

Highest CCR → 0

D. Data Sovergenity and community voice:

$$\text{Algorithmic Green Sovergenity Score (AGSS)} = \frac{DO+BV+AR+LOI}{4}$$

Where

DO = Data ownership

BV = Binding veto

AR = Algorithmic recourse

LOI = Local oncology inclusion

All the scores are from 0 to 100.

Data Sovergenity Use: Data Sovergenity (DS) = Data ownership (DO)

Community voice use:

$$\text{Community voice (CV)} = \frac{BV+AR+LOI}{3}$$

E. Verified Impact (VI):

$$CCR = \frac{\text{Model emissions}}{\text{Verified abatement}}$$

Therefore, verified abatement is present in CCR, but verified impact itself isn't a separate table 5 is

$$\text{Verified Impact (VI)} = \frac{100*\text{verified abatement}}{\text{Target/reference abatement}}$$

Illustrative scoring of the six governance dimensions

Dimension	Status quo	GFJM governed	Scoring basis
Community Voice	35	85	AGSS governance components
Green-spread Parity	35	85	Normalized inverse GMG
Credit Access Parity	30	85	100(1-CRI)
Verified Impact	40	80	Verified-abatement performance
Data Sovereignty	25	90	AGSS data-ownership component
Compute Frugality	30	85	Normalized inverse CCR

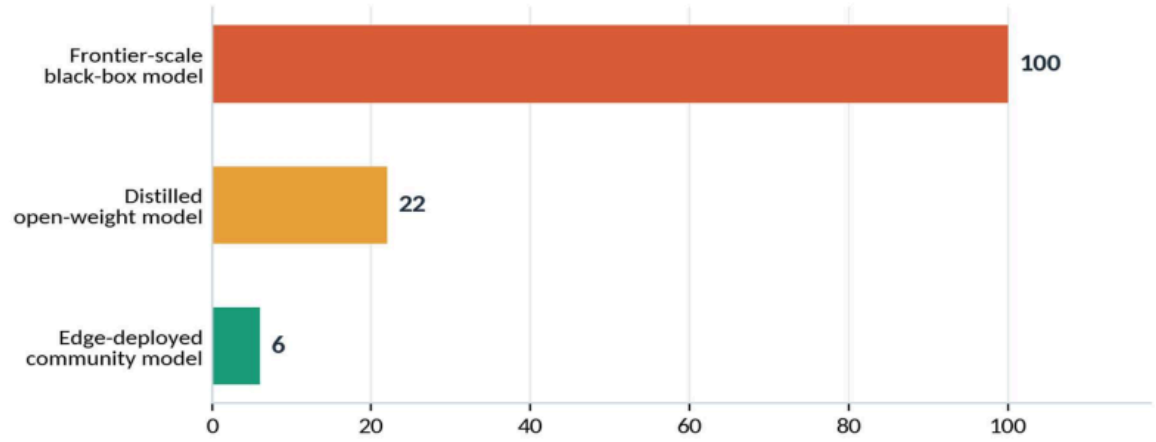


Figure 5 (carbon per 1,000 decisions by model)

Sources: Strubell et al. (2019), Luccioni et al. (2023), Patterson et al. (2021), the ML.ENERGY leaderboard, IEA's *Energy and AI* report, and de Vries-Gao (2025).

Furthermore, these findings suggest that the two dimensions represent areas where technical fairness fixes rarely provide meaningful change. Given that technical approaches demonstrate limitations, the results might indicate that governance mechanisms could address these critical gaps more effectively.

6. Case Scenarios: Redesigning Green-Fintech Systems

The scenarios appear to provide that illustrative design examples rather than observed implementations. Moreover, the scenarios may demonstrate that potential governance redesign opportunities could exist and might require empirical evaluation.

Scenario	Status quo	GFJM intervention
A. Green credit for smallholders and MSMEs	Opaque climate-risk scores widen spreads for exposed farms;	Community-defined vulnerability and resilience indexes; green data

	alternative data is harvested without return	trust with a reciprocity ledger; recourse for contested scores
Carbon-credit markets and digital MRV	Satellite and AI verification is proprietary; forest communities supply ground truth but receive little of the credit value	Community oracles co-verify outputs; binding benefit-sharing through the trust; published model carbon budget
Parametric climate insurance	Index triggers calibrated on urban or insured-asset data miss rural loss patterns; payouts are slow or denied	Pluriversal trigger design using local ontologies of loss; community audit before deployment; deterministic, open triggers

Table 6. Redesigning three green-fintech systems.

7. Research Agenda and Limitations:

Operationalizing the four indices might suggest that risk-adjusted lending data, lifecycle assessments of model stacks and community-validated sovereignty tests appear necessary. Given that frugal models may sacrifice some predictive accuracy, the appropriate accuracy-footprint frontier appears to demonstrate that normative choice remains necessary for affected communities, not only for modelers. Nevertheless, the governance risk could suggest that community bodies appear vulnerable to capture. Governance risk shows community bodies captured. Additionally, rotating membership, independent technical support and transparent funding may indicate that design necessities could exist. In light of these limitations, the quantitative evidence here might suggest that illustrative examples appear present, and the propositions remain untested.

In terms of justice, the framework has a strongly normative orientation towards notions of distributive and procedural justice which may not be acceptable to policymakers and financial institutions. Further, the community governance structure itself may be susceptible to elite capture, unequal representation, and institutional conflicts. In the current study GFJM has not been empirically evaluated in jurisdictions.

8. Conclusion: Towards Green Finance

The findings suggest that decarbonisation cannot be financed on terms that may deepen the exposure of those who contributed least to the crisis. Moreover, the evidence indicates that retrofitting green-fintech models with superficial fairness constraints might not interrupt the double margin. Nevertheless, the Green Fintech Justice Matrix could provide an alternative approach: the results demonstrate that treating marginality as enclosure appears relevant. In light of these findings, the data suggest that pairing every pillar with a mechanism and a measurable index could establish key outcomes. Results show participation strengthens accountability. However, the study may indicate that community participation and accountability mechanisms within climate-finance governance systems appear important. Furthermore, these systems could provide the evidence on which green value is certified. Thus, the findings suggest that if the algorithmic margin is reclaimed, green fintech might become an instrument of collective

adaptation. Given that extraction represents another frontier, the evidence may demonstrate that this approach could establish a significant alternative.

Conflict of Interest Statement

Authors declare no competing interests influence research. Nevertheless, the authors may indicate that no financial interests could inappropriately bias the findings presented in this paper. Given that transparency appears critical, the disclosures might suggest that the research maintains integrity. Additionally, the authors could demonstrate that personal interests appear absent from the study.

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